A Four-by-Four Obstruction to Polynomial Removal for Ordered Binary Matrices

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October 11, 2026

Abstract

We exhibit a fixed 4×4 binary matrix with no polynomial induced ordered removal bound. At depth h, write $m = 2^h$. Our explicit host has order $(14h + 2)m$, exactly $2m^6 + \frac{1}{2}(m-2)m^5$ induced copies, and exact editing distance m^2 cells. Consequently its normalized edit distance is $\Theta(h^{-2})$, while its copy density is $\Theta(4^{-h}h^{-8})$. The lower bound holds against arbitrary cell edits by an injective family of m^2 protected selections; an arithmetic classification identifies every copy and certifies that a matching canonical repair creates none. The original-copy transversal number is exactly m, establishing an unbounded gap between hitting existing copies and eliminating all copies. The identities and the all-depth obstruction are formalized in Lean, with complete certificate sources and reproducible verification procedures.

1 Introduction and main results

Let H be a binary $k \times k$ matrix and let A be a binary $n \times n$ matrix. An induced ordered copy of H in A is a pair of strictly increasing k-tuples of row and column indices whose induced submatrix agrees with H in every entry. Write $N_H(A)$ for the number of such copies. Let $D_H(A)$ be the minimum number of cell changes that turn A into an H-free matrix, allowing both $0 \to 1$ and $1 \to 0$ changes, and write $\mathrm{dist}_H(A) = D_H(A)/n^2$. A polynomial induced removal bound for H is an inequality of the form

$$N_H(A) \geq c\, \mathrm{dist}_H(A)^C n^{2k} \tag{1}$$

with constants $c, C > 0$ independent of A and n. Ordered removal lemmas and their property-testing consequences are studied in [2, 3]. The earlier construction of [6] exhibited a 66×66 obstruction to polynomial removal. Here the forbidden pattern is 4×4 and remains fixed at every depth. We determine its exact copy count and unrestricted editing distance, and identify the original-copy transversal number. Unlike an original-copy hitting set, an editing set must leave a matrix free of all copies, including those created by the edits (see [3]). The smaller forcing frame and its terminal classification provide the principal new quantitative results.

We consider the pattern

$$H_4 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} J_2 - I_2 & I_2 \\ I_2 & P \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \tag{2}$$

The rows of H_4 are pairwise distinct, as are its columns. Hence physical copies can be counted by weighted block tuples (Lemma 2.2).

Theorem 1.1 (Main theorem). *For every integer $h \geq 1$, put $m = 2^h$ and $n_h = (14h+2)m$. Let A_h be the binary matrix constructed in Section 2. Then*

$$D_{H_4}(A_h) = m^2, \qquad \operatorname{dist}_{H_4}(A_h) = \frac{1}{(14h+2)^2}, \qquad N_{H_4}(A_h) = 2m^6 + \frac{m-2}{2}m^5. \tag{3}$$

Moreover, the canonical repair B_h is H_4-free, and every induced ordered copy in A_h belongs to exactly one of the two terminal families described in Section 4.

Corollary 1.2 (Failure of polynomial removal). *The pattern H_4 admits no polynomial induced ordered binary matrix removal bound. More precisely, if*

$$\epsilon_h = (14h+2)^{-2}, \qquad \delta_h = \frac{N_{H_4}(A_h)}{n_h^8},$$

then

$$n_h \longrightarrow \infty, \qquad \epsilon_h \longrightarrow 0, \qquad \frac{\delta_h}{\epsilon_h^C} \longrightarrow 0 \quad \text{for every fixed } C \in \mathbb{R}.$$

Proof. The theorem gives, with $d_h = 14h+2$,

$$\delta_h = \frac{5/2 - 2^{-h}}{d_h^8 4^h} \leq \frac{3}{4^h}.$$

Since $d_h \to \infty$ and $n_h = d_h 2^h$, we have $n_h \to \infty$ and $\epsilon_h \to 0$. Fix $C \in \mathbb{R}$ and choose a natural number $k > C$. Since $0 < \epsilon_h \leq 1$, we have

$$0 \leq \frac{\delta_h}{\epsilon_h^C} \leq \frac{\delta_h}{\epsilon_h^k} \leq \frac{3d_h^{2k}}{4^h} \longrightarrow 0.$$

The last limit is the standard comparison of polynomial growth with geometric growth. Given $c, C > 0$, the inequality $\delta_h < c\epsilon_h^C$ therefore holds for all sufficiently large h, which contradicts (1) along the family A_h. □

The proof separates the unrestricted editing problem from the enumeration of copies. Section 2 defines the block-constant host and its canonical repair. Section 3 constructs m^2 protected selections and proves the lower bound for arbitrary edits. The terminal classification and repaired freeness appear in Section 4; Section 5 then closes the main theorem by counting the physical copies with their block multiplicities. The exact transversal number and the quantitative density profile follow in subsequent sections. Structural details, complementary proofs, arithmetic certificates and verification instructions are collected in the appendices. The tree-propagation mechanism follows the general approach of [6]; the forcing frame and the exact classifications here are specific to H_4.

2 The host construction

Fix $h \geq 1$ and write $m = 2^h$. For each internal level $0 \leq i < h$ and node $0 \leq p < 2^i$, introduce row blocks $R_i^{\pm}(p)$ and column blocks $C_i^{\pm}(p)$, each of size 2^{h-i}. At level h the two signs are identified: a leaf label $p < m$ determines one row position $R_h(p)$ and one column position $C_h(p)$. Finally, add an m-element dummy block on each axis, denoted by R_* and C_*.

The variable blocks form binary trees. On the row axis the minus block at a node precedes the two child subtrees and the plus block follows them; on the column axis these two positions are reversed. In both trees the left child subtree precedes the right child subtree. The row dummy block follows the tree, whereas the column dummy block precedes it.

For each $1 \le i \le h$ we add six marker modes

$$V_i^+,\ V_i^-,\ W_{i,0}^+,\ W_{i,1}^+,\ W_{i,0}^-,\ W_{i,1}^-.$$

Each mode t has marker row blocks $A_1(t), A_2(t)$ and marker column blocks $B_1(t), B_2(t)$, all of size m. The two marker groups interact with the following non-marker classes:

Mode	Row role 1	Row role 2	Column role 1	Column role 2
V_i^+	R_i^+	R_{i-1}^+	C_*	C_{i-1}^+
V_i^-	R_{i-1}^-	R_i^-	C_*	C_{i-1}^-
$W_{i,b}^+$	$R_{i,b}^+$	R_*	C_{i-1}^+	$C_{i,b}^+$
$W_{i,b}^-$	$R_{i,b}^-$	R_*	$C_{i,b}^-$	C_{i-1}^-

Here $R_i^\pm$ and $C_i^\pm$ denote the unions of all blocks at the indicated level and sign; a subscript b restricts the union to nodes of parity b. At level h, a leaf belongs to the class determined by its parity for either sign.

All marker blocks precede the variable and dummy blocks. The marker prefix starts with

$$W_{1,0}^-, W_{1,1}^-, \dots, W_{h,0}^-, W_{h,1}^-;$$

the remaining modes occur in increasing level order as $V_i^+, V_i^-, W_{i,0}^+, W_{i,1}^+$. Within each mode, group 1 precedes group 2. These orders are used separately on the row and column axes.

The entry rule is constant on every product of two blocks. Two marker blocks have entry one exactly when their modes agree and their group indices differ. The support of a group-u marker row on non-marker columns is exactly column role u of its mode; the support of a group-u marker column on non-marker rows is exactly row role u. Every dummy–non-marker entry is one. The remaining variable–variable entries are

$$\begin{aligned}
A_h(R_i^+(p), C_i^+(q)) &= \mathbf{1}[p \le q] && (i < h),\\
A_h(R_i^-(p), C_i^-(q)) &= \mathbf{1}[p < q] && (i < h),\\
A_h(R_h(p), C_h(q)) &= \mathbf{1}[p \le q], && \\
A_h(R_i^+(p), C_{i-1}^+(q)) &= \mathbf{1}[\lfloor p/2 \rfloor \le q], &&\\
A_h(R_i^-(p), C_{i-1}^-(q)) &= \mathbf{1}[\lfloor p/2 \rfloor < q]. &&
\end{aligned} \tag{4}$$

In the last two identities $1 \le i \le h$, and a leaf row at $i = h$ is treated as either signed row. The indicator $\mathbf{1}[S]$ is one when S holds and zero otherwise. All other variable–variable entries are zero.

The canonical repair B_h changes precisely the entries in

$$R_* \times \{C_h(p) : 0 \le p < m\}$$

from one to zero and leaves all other entries unchanged.

Lemma 2.1 (Physical size and order)**.** *Each axis contains exactly* $(14h + 2)m$ *positions. After ordering the positions within every block by their offsets, the resulting block order extends to a strict total order. Consequently each axis is order-isomorphic to* $\{0, \dots, n_h - 1\}$ *with its usual order.*

Proof. The marker blocks contribute $6h \cdot 2m = 12hm$ positions. At each internal depth, the 2^i nodes on either sign have multiplicity 2^{h-i}, hence contribute m positions; the two signs contribute $2hm$ in total. The leaves and the dummy block contribute m positions each. This gives $(14h + 2)m$. Ordering offsets increasingly inside each block gives the required total order and the order-preserving bijections. We use these bijections to regard A_h and B_h as numerically indexed matrices. □

Lemma 2.2 (Block quotient and multiplicity)**.** *An induced ordered copy of H_4 in either host uses four distinct row blocks and four distinct column blocks. The ordered block tuples whose entry table equals H_4 are precisely the block projections of physical copies. A tuple has exactly the product of its eight block sizes as physical realizations.*

Proof. Two rows from one block agree on every column, so their signatures in a copy would be equal. The distinct rows of H_4 exclude this possibility; the distinct columns give the same conclusion on the other axis. Each block is an interval in its axis order. Conversely, choosing one position from each block of an ordered tuple preserves its order and all entries, because the entry rules are block-constant. The choices are independent and have the stated product as their number. □

3 The edit-distance lower bound

A *marker frame* is an ordered choice of four rows and four columns whose twelve entries outside the lower-right 2×2 block agree with H_4. Its lower-right block is a copy of P exactly when the frame is a copy of H_4.

3.1 Protected selections

Index a family of selections by pairs $(a, b) \in \{0, \dots, m-1\}^2$. Set $z = a \oplus b$, with $\oplus$ denoting bitwise XOR, and at depth $d \le h$ select the node

$$p_d = \left\lfloor \frac{z}{2^{h-d}} \right\rfloor .$$

Then $p_0 = 0$, $p_h = z$ and $p_{d-1} = \lfloor p_d/2 \rfloor$ for $1 \le d \le h$. In a signed row block on this path choose the offset $a \bmod 2^{h-d}$; in the corresponding signed column block choose $b \bmod 2^{h-d}$. In each marker block and in the dummy block use offset a on rows and offset b on columns. At depth h, the two signed paths meet at the same leaf on each axis.

Call a cell in the product of the selected rows and columns protected when at least one endpoint is a marker or dummy position, or when both endpoints are root variable positions. Thus every selected frame contributes its twelve prescribed entries, together with the dummy incidences and root entries used in the propagation argument.

Lemma 3.1 (Injectivity of protection)**.** *Every physical cell is protected by at most one of the m^2 selections.*

Proof. If both endpoints are marker or dummy positions, their offsets recover (a, b). Suppose instead that the row endpoint is a variable position at depth d and the column endpoint is a marker or dummy position. The column offset recovers b. The row node recovers $\lfloor z/2^{h-d} \rfloor$, while the row offset recovers $a \bmod 2^{h-d}$. Since XOR commutes with truncation of the lower $h - d$ bits,

$$\left\lfloor \frac{a}{2^{h-d}} \right\rfloor = p_d \oplus \left\lfloor \frac{b}{2^{h-d}} \right\rfloor .$$

The quotient and remainder therefore determine a. When the row is a marker or dummy and the column is variable, the row offset gives a, the column node gives p_d, and the identity $\lfloor b/2^{h-d} \rfloor = p_d \oplus \lfloor a/2^{h-d} \rfloor$ recovers the quotient of b; its column offset supplies the remainder. A protected cell with two variable endpoints occurs at the roots, where its two offsets directly recover (a, b). At a leaf $d = h$, the offset is zero and $p_h = z$, so the mixed endpoint cases reduce to $a = z \oplus b$ or $b = z \oplus a$, respectively. □

Corollary 3.2 (An untouched protected selection)**.** *If fewer than m^2 cells are changed, some selection has all its protected cells unchanged.*

Proof. By Lemma 3.1, each changed cell affects at most one selection. Fewer than m^2 changed cells can therefore affect fewer than m^2 selections. □

3.2 Propagation along a protected path

Fix a selection whose protected cells remain unchanged, and let D be an H_4-free matrix obtained from A_h by editing cells. Let x_i and y_i be the selected plus and minus diagonal entries at depth i, respectively. Thus $x_i = D(r_i^+, c_i^+)$ and $y_i = D(r_i^-, c_i^-)$, where lower-case letters denote the selected positions in the corresponding blocks. The protected root entries give $x_0 = 1$ and $y_0 = 0$. Put $\alpha_i = D(r_i^+, c_{i-1}^+)$ and $\beta_i = D(r_i^-, c_{i-1}^-)$. The four selected marker frames have bodies

Mode	Body	Consequence of H_4-freeness
V_i^+	$\left(\begin{smallmatrix} 1 & \alpha_i \\ 1 & x_{i-1} \end{smallmatrix}\right)$	$x_{i-1} = 1 \implies \alpha_i = 1$
V_i^-	$\left(\begin{smallmatrix} 1 & y_{i-1} \\ 1 & \beta_i \end{smallmatrix}\right)$	$y_{i-1} = 0 \implies \beta_i = 0$
$W^+_{i,p_i \bmod 2}$	$\left(\begin{smallmatrix} \alpha_i & x_i \\ 1 & 1 \end{smallmatrix}\right)$	$\alpha_i = 1 \implies x_i = 1$
$W^-_{i,p_i \bmod 2}$	$\left(\begin{smallmatrix} y_i & \beta_i \\ 1 & 1 \end{smallmatrix}\right)$	$\beta_i = 0 \implies y_i = 0$

The ordered non-marker rows and columns of these frames are

Mode	First row	Second row	First column	Second column
V_i^+	$R_i^+(p_i)$	$R_{i-1}^+(p_{i-1})$	C_*	$C_{i-1}^+(p_{i-1})$
V_i^-	$R_{i-1}^-(p_{i-1})$	$R_i^-(p_i)$	C_*	$C_{i-1}^-(p_{i-1})$
$W^+_{i,p_i \bmod 2}$	$R_i^+(p_i)$	R_*	$C_{i-1}^+(p_{i-1})$	$C_i^+(p_i)$
$W^-_{i,p_i \bmod 2}$	$R_i^-(p_i)$	R_*	$C_i^-(p_i)$	$C_{i-1}^-(p_{i-1})$

Each pair is preceded by the two marker groups of its mode. In the row tree a plus descendant precedes its plus ancestor, whereas a minus ancestor precedes its minus descendant. In the column tree these relations are reversed. The column dummy precedes all variables and the row dummy follows them. These four observations give the strict orders in every row of the table, including the terminal level. The twelve marker entries and all displayed dummy entries are protected. Induction on i now gives

$$x_{i-1} = 1 \implies \alpha_i = 1 \implies x_i = 1, \qquad y_{i-1} = 0 \implies \beta_i = 0 \implies y_i = 0.$$

Starting from $x_0 = 1$ and $y_0 = 0$, these implications give $x_i = 1$ and $y_i = 0$ for every $i \le h$. At depth h the signed row positions are identified, and so are the signed column positions. Thus x_h and y_h are the same physical cell, which would have to equal both 1 and 0.

Proposition 3.3 (Arbitrary-edit lower bound)**.** *Every binary H_4-free matrix obtained from A_h by cell edits differs from A_h in at least m^2 cells.*

Proof. If fewer than m^2 cells were changed, Corollary 3.2 would provide an untouched protected selection. The propagation argument above would then produce a contradiction. □

4 Terminal classification and canonical repair

For a leaf label $p < m$, put $t = W^-_{h,p \bmod 2}$. The two terminal forms use the common row tuple

$$A_1(t), \quad A_2(t), \quad R_h(p), \quad R_*.$$

Their column tuples are

$$\begin{aligned} \text{I}: &\quad B_1(t),\ B_2(t),\ C_h(p),\ C^-_{h-1}(\lfloor p/2 \rfloor), \\ \text{II}: &\quad B_1(t),\ C_h(q),\ C_h(p), C^-_{h-1}(\lfloor p/2 \rfloor), \quad q < p,\ q \equiv p \pmod 2. \end{aligned}$$

The row tuple is ordered because the marker groups precede the leaf and the dummy row follows it. The leaf column $C_h(p)$ precedes its minus parent in the column tree. For form II, the conditions $q < p$ and $q \equiv p \pmod 2$ imply $\lfloor q/2 \rfloor < \lfloor p/2 \rfloor$, so $C_h(q)$ also precedes that parent and precedes $C_h(p)$. Thus both displayed tuples are ordered. The entry table gives the four row vectors

$$(0,1,1,0), \quad (1,0,0,1), \quad (1,0,1,0), \quad (0,1,1,1).$$

In form II, the same parity makes $C_h(q)$ a column-role-1 leaf for t, and $q < p$ makes its entry against $R_h(p)$ zero. These give the same second-column signature as the second marker in form I.

Lemma 4.1 (Exhaustive terminal classification)**.** *A valid ordered block tuple forms an induced ordered copy of H_4 in A_h if and only if it has terminal form I or terminal form II.*

Proof. Let $r_1 < \cdots < r_4$ and $c_1 < \cdots < c_4$ be the selected block types. By Lemma G.1, their integer encodings satisfy the original copy predicate $\mathcal{C}^0_h$. The first implication of Lemma G.2 places them in the integer terminal family.

The terminal predicate identifies the first three rows as two level-h W^- marker rows followed by a terminal leaf, the fourth row as the dummy, and the third column as the same terminal leaf. The fourth column is the minus parent of that leaf. The entry between the leaf row and the first marker column is $(H_4)_{31} = 1$. By the role table it forces the marker parity to equal p mod 2. The second column is then either the second marker of the same mode or an earlier terminal leaf with the same parity. Physical block validity restores $0 \le p < m$; in the second alternative it also gives $0 \le q < p$. These are precisely forms I and II. The leaf row determines p, and the kind of the second column distinguishes the alternatives. In form II that column also determines q, proving uniqueness of the parameters.

The ordered tuples and their entries were checked above, giving the converse. □

Lemma 4.2 (Freeness of the canonical repair)**.** *For every $h \ge 1$, the repaired matrix B_h is H_4-free.*

Proof. Suppose a copy exists in B_h. Lemma 2.2 gives an ordered block tuple realizing it, and Lemma G.1 sends that tuple to a witness for $\mathcal{C}^1_h$. The second implication of Lemma G.2 excludes such a witness. This argument applies to the entire repaired entry table. The bijections in Lemma 2.1 identify the conclusion with freeness in the usual numerical matrix. □

5 Exact count and distance

By Lemma 2.2, each terminal tuple contributes the product of its block multiplicities. For form I this product is $2m^5$: the four marker blocks and the dummy row each have size m, the parent column has size two, and the leaf row and column each have size one. For form II there are three marker blocks, a dummy row, three leaf positions and a parent column, giving $2m^4$. The contributions are summarized by

Family	Number of parameters	Copies per parameter	Contribution
I	m	$2m^5$	$2m^6$
II	$m(m-2)/4$	$2m^4$	$(m-2)m^5/2$

Indeed, the admissible pairs for form II are the strictly ordered pairs within each of the two parity classes, each of size $m/2$. Hence

$$\#\{(q,p) : 0 \le q < p < m,\ q \equiv p \pmod 2\} = 2\binom{m/2}{2} = \frac{m(m-2)}{4}.$$

For $h = 1$, $m = 2$ and the second family is empty, as the formula gives. The two families are disjoint and their parameters are uniquely recovered from the selected blocks. The classification lemma therefore yields

$$N_{H_4}(A_h) = 2m^6 + \frac{m-2}{2}m^5.$$

Proof of Theorem 1.1. Lemma 2.1 gives $n_h = (14h+2)m$, and the preceding count gives the last identity in (3). The canonical repair changes exactly the m^2 cells in $R_* \times \{C_h(p) : p < m\}$ and is H_4-free. Proposition 3.3 shows that no repair can use fewer cells. Thus $D_{H_4}(A_h) = m^2$, and dividing by n_h^2 gives the stated normalized distance. The two remaining assertions are Lemmas 4.1 and 4.2. □

6 Hitting original copies versus eliminating all copies

Corollary 6.1 (Exact transversal number)**.** *For every $h \ge 1$, with $m = 2^h$, the host satisfies*

$$\tau_{H_4}(A_h) = m < m^2 = D_{H_4}(A_h).$$

In particular, the ratio of unrestricted editing distance to the minimum number of entries hitting all original copies is exactly m.

Proof. Let

$$\Delta_h = \{(R_h(p), C_h(p)) : 0 \le p < m\}.$$

Both terminal families contain the same-leaf diagonal $(R_h(p), C_h(p))$ as their third-row, third-column entry. The leaf blocks are singletons, so this is one fixed physical cell for every realization of the corresponding terminal tuple. Lemma 4.1 therefore gives $\tau_{H_4}(A_h) \le |\Delta_h| = m$.

For the reverse inequality, use offsets numbered from zero. For each $p < m$, put $t_p = W^-_{h,p \bmod 2}$ and choose a type-I copy Q_p whose two marker rows, two marker columns and dummy row all have offset p in their respective blocks. Use the singleton leaf row and column with label p, and use offset $p \bmod 2$ in the parent column block $C^-_{h-1}(\lfloor p/2 \rfloor)$. These choices are valid: the marker and dummy blocks have size m, and each parent block has size two. They realize an ordered copy by the type-I description and Lemma 2.2.

The copies Q_p are pairwise disjoint on each axis. Marker positions of different modes or groups lie in different blocks; within any one such block, different labels p give different offsets. The dummy row offsets and leaf labels are also distinct. Two selected parent columns could agree only if both $\lfloor p/2 \rfloor$ and $p \bmod 2$ agree, which determines p. The marker, leaf, dummy and parent categories used on a given axis are mutually distinct. Thus the m copies are, in particular, pairwise cell-disjoint. Every set hitting all original copies must meet each Q_p in a different cell, and hence has size at least m. Combining the bounds proves the transversal identity; the editing identity is Theorem 1.1, and $m \ge 2$ gives strictness. □

The strict gap between hitting original copies and producing a free matrix sharpens the distinction already developed in [6]. A separate Lean supplement formalizes the exact transversal identity, using terminal exhaustion from the primary classification theorem.

7 Quantitative consequences of the exact formulas

The main theorem determines the obstruction at three scales: tree depth h, matrix order n_h, and normalized edit distance ϵ_h. Explicit changes of scale sharpen the comparison with polynomial removal inequalities.

Proposition 7.1 (Stretched-exponential copy-density profile)**.** *At the exact values* $\epsilon_h = (14h + 2)^{-2}$*, the copy density is*

$$\delta_h = \left(\frac{5}{2} - 2^{-h}\right)\epsilon_h^4 \exp\left[-\frac{\log 4}{14}(\epsilon_h^{-1/2} - 2)\right]. \tag{5}$$

In particular, the largest valid lower-bound function for the individual pattern H_4 *cannot exceed the right-hand side along this sequence of edit parameters.*

Proof. By Theorem 1.1, $\delta_h = (5/2 - 2^{-h})/((14h + 2)^8 4^h)$. The relation $\epsilon_h^{-1/2} = 14h + 2$ gives $h = (\epsilon_h^{-1/2} - 2)/14$. Substitution yields (5). For any function $g(\epsilon)$ satisfying $N_{H_4}(A) \geq g(\mathrm{dist}_{H_4}(A))n^8$ for all finite A, applying the bound to A_h gives $g(\epsilon_h) \leq \delta_h$. □

Proposition 7.2 (Profile by matrix order)**.** *As* $h \to \infty$*, the constructed orders and copy counts satisfy*

$$\epsilon_h = \Theta((\log n_h)^{-2}), \qquad \delta_h = \Theta\left(\frac{1}{n_h^2(\log n_h)^6}\right), \qquad N_{H_4}(A_h) = \Theta\left(\frac{n_h^6}{(\log n_h)^6}\right).$$

Proof. Writing $d_h = 14h+2$ and $m = 2^h$ gives $\log n_h = h \log 2 + \log d_h = \Theta(h)$, so $d_h = \Theta(\log n_h)$ and $m = n_h/d_h$. The exact copy formula may be rearranged as

$$N_{H_4}(A_h) = \left(\frac{5}{2} - \frac{1}{m}\right)\frac{n_h^6}{d_h^6}.$$

The coefficient lies in $[2, 5/2)$ because $m \geq 2$. Dividing by n_h^8 and using the distance identity gives the three assertions. □

7.1 A small-depth consistency table

The exact formulas provide convenient integer-valued checks of the terminal enumeration. They require no asymptotic approximation.

h	m	n_h	family I	family II	$N_{H_4}(A_h)$
1	2	32	128	0	128
2	4	120	8192	1024	9216
3	8	352	524288	98304	622592
4	16	928	33554432	7340032	40894464

The two contributions follow by substituting $m = 2^h$ into the form-I and form-II expressions. At depth one, family II is absent; at higher depths, its share rises toward one fifth of the total. The table illustrates why a search that detects only the single explicit propagation family cannot certify the complete copy count.

A Background and interpretation

The quantitative issue for ordered matrices is the dependence on the edit parameter. The row and column orders are independent: an occurrence of a pattern $F \in \{0,1\}^{s\times t}$ is a pair of increasing injections $[s] \hookrightarrow [n]$ and $[t] \hookrightarrow [n]$, and all st entries, including the zeros, are prescribed. The corresponding hereditary property consists of matrices avoiding F as an induced ordered submatrix. A removal bound supplies a lower bound on the number of occurrences whenever one must change a prescribed fraction of all entries to obtain a member of this hereditary property.

For an individual fixed F, a polynomial bound means the existence of $c_F, C_F > 0$ such that the right side of (1) holds for *every* size and *every* matrix. Failure is witnessed by one fixed pattern and a sequence of arbitrarily large hosts. The quantifiers matter: choosing a different forbidden pattern as the accuracy parameter decreases would address a different question. The present family uses the same H_4 at every depth; the pattern is independent of h, m and n_h.

We normalize occurrences by n^{2k}, as in (1), and count increasing injections, so a copy has no additional factorial multiplicity. For a fixed k, using $\binom{n}{k}^2$ instead changes the density only by a factor bounded above and below by positive constants for $n \geq k$. Thus the failure of a polynomial bound is independent of these two common conventions.

Alon, Ben-Eliezer and Fischer [2] proved qualitative testing results for hereditary properties of ordered graphs and matrices. Alon and Ben-Eliezer [3] obtained a polynomial dependence for a *hitting* formulation: a small set of entries may intersect all copies present in the original matrix. Hitting the original copies need not produce a free matrix after entry changes, because a changed zero or one can complete a new induced copy. The relevant full removal problem minimizes changes over the resulting, genuinely F-free matrices. This distinction is particularly significant for an induced pattern, whose constraints include prescribed zeros as well as ones. Alon and Ben-Eliezer also establish a full polynomial removal lemma for every fixed finite family of binary matrices closed under row permutations [3, Theorem 1.5], and hence, by transposition, for families closed under column permutations. The singleton $\{H_4\}$ has neither closure property, so this positive result does not apply to the present obstruction. Gishboliner and Shapira [5, Conjecture 4.5] formulate the full polynomial removal conjecture for a single fixed ordered binary matrix. Towsnerś ordered-hypergraph removal lemma provides a further point of comparison at the level of qualitative removal [4]. Earlier work on unordered bipartite matrices and conditional regularity provided the polynomial-query comparison [1]; the ordering requirement removes the row-and-column permutation freedom used in that setting.

The 2026 construction of [6] gave a fixed 66×66 binary obstruction. Its tree-based construction showed that forcing gadgets can make edit distance persist over successive levels, while an order constraint can confine complete copies to terminal locations. Our pattern (2) has four rows and four columns; its top-left 2×2 anti-diagonal marker and the two identity coupling blocks provide a smaller frame for the same kind of propagation. The sharper size is accompanied here by exact distance and copy-count identities. We isolate the roles of the order, the two signs, and terminal identification in subsequent sections.

The tree-based propagation and terminal-sign identification build on [6]. The additional results here are the explicit 4×4 pattern, its exact physical copy count, its optimal unrestricted repair and the formal certification of these identities.

A.1 Three quantities that must be kept separate

Let $\mathcal{C}_H(A)$ be the set of increasing row/column tuples realizing H. Define the *transversal number*

$$\tau_H(A) = \min\{|S| : S \subseteq [n] \times [n],\ S \cap \mathrm{cells}(Q) \neq \varnothing\ \forall Q \in \mathcal{C}_H(A)\}.$$

Every edit set turning A into an H-free matrix intersects every copy originally in A; consequently

$$\tau_H(A) \le D_H(A). \tag{6}$$

Equality does not follow from the definition: a hitting set has no obligation to avoid copies created by its own edits. For our host, the protected-selection argument establishes the stronger lower bound $D_{H_4}(A_h) \ge m^2$ *for all resulting matrices.* The canonical repair gives a matching upper bound by an explicit final H_4-free matrix. Neither step is supplied merely by counting original occurrences.

Indeed, the terminal classification below shows a concrete gap between the two quantities. The m leaf-diagonal cells $\{(R_h(p), C_h(p)) : 0 \le p < m\}$ intersect every original copy, so $\tau_{H_4}(A_h) \le m < m^2 = D_{H_4}(A_h)$. Corollary 6.1 proves that the hitting bound is exact. Every terminal copy contains its same-leaf diagonal as its third-row, third-column cell. This is the hitting-versus-removal phenomenon emphasized in [6], here retained by the smaller pattern; it is not a new general separation principle.

It is also useful to distinguish physical copies from type-level copies. In a block-constant host, quotient rows and columns encode signatures and ordering, whereas block multiplicities control the physical number of realizations. The quotient alone has no absolute copy count unless its tuples are weighted by those multiplicities. This separation is essential in the terminal enumeration.

A.2 The four-by-four pattern as a forcing frame

Partition H_4 after the second row and the second column. The four component blocks are respectively $J_2 - I_2$, I_2, I_2, and the asymmetric body P. Thus two marker rows and columns fix twelve entries of a potential induced copy; the residual body has only four entries. If the residual body has the form $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, forbidding H_4 imposes the single constraint

$$(a, b, c, d) \ne (1, 0, 1, 1).$$

The construction uses this exact negative constraint in four orientations. Two orientations push a value from level $i - 1$ to a cross-level cell, and two push it from that cell to the diagonal at level i. The plus chain propagates the value one; the minus chain propagates zero. Their terminal positions coincide. The gadget constraints therefore become inconsistent unless an entry required by a selected frame has been changed. The XOR choice of physical representatives makes those required entries disjoint across m^2 choices.

B Order and block structure in detail

This section records elementary consequences of the host definition that are useful when interpreting the certificate classification. The binary word of a node (d, p) consists of the first d bits in the length-h binary expansion of an appropriate leaf label. An internal signed node is represented by the corresponding word, together with its sign. The leaf at depth h has a single type.

Lemma B.1 (Tree comparison by words)**.** *Take two distinct variable blocks of the same axis. If their node words have an incomparable prefix, their order is determined by the first bit on which they differ. If one word is a proper prefix of the other, the ancestor precedes the descendant precisely when its sign is the opening sign for that axis. The opening signs are minus for rows and plus for columns. If the two words agree, the blocks are the two signs at one internal node, and the opening block precedes the closing block.*

Proof. At any internal vertex the recursive row list has the form

$$R_d^-(p), \quad \text{left subtree}, \quad \text{right subtree}, \quad R_d^+(p),$$

whereas the column list exchanges the two signed blocks. If the words separate into the two different child subtrees, the left subtree precedes the right regardless of the signs within either. If one word names an ancestor of the other, the ancestor's opening block occurs before the descendant's entire subtree and its closing block after it. Finally, equal words can name distinct blocks only at an internal node with opposite signs; the displayed recursive lists place the opening sign before the closing sign. These three cases exhaust pairs of distinct variable blocks. □

Proposition B.2 (A uniform quotient principle)**.** *Let F be an $s \times t$ binary matrix with no repeated row signature and no repeated column signature. Let M be any finite block-constant ordered binary matrix whose row blocks and column blocks are nonempty consecutive intervals. Then*

$$N_F(M) = \sum_{(r,c)\in\mathcal{Q}_F(M)} \prod_{i=1}^{s} |R_{r_i}| \prod_{j=1}^{t} |C_{c_j}|,$$

where $\mathcal{Q}_F(M)$ consists of the increasing row-block and column-block tuples realizing the entries of F on the quotient. Moreover, M is F-free if and only if its quotient has no such tuple.

Proof. If two chosen physical rows belonged to the same block, then they would have identical entries on every chosen column, contradicting the distinctness of the corresponding rows of F. The column argument is symmetric. It follows that each physical copy projects to a unique quotient tuple with strictly increasing block indices. Conversely, any tuple of quotient blocks realizing F admits a physical copy by choosing one position from each participating block. Every such choice preserves the order, because the blocks are intervals; it also preserves all entries, because entries are block-constant. The choices are independent and their number is the product of the participating block sizes. Summing over the quotient tuples gives the identity and its freeness consequence. □

There are two independent hypotheses in Proposition B.2. The consecutive-interval requirement is responsible for preservation of the ordered embedding, while distinct row and column signatures ensure that the selected type tuples are strictly increasing. For patterns with repeated signatures one may have several physical rows in the same type, and a multiplicity formula requires binomial factors and within-block order. The simple product formula suffices for H_4 and for the general twin-free patterns considered in Appendix H.

B.1 Exact support signatures

For a mode t and its two marker groups, write $\rho_1(t), \rho_2(t)$ for row-role predicates and $\kappa_1(t), \kappa_2(t)$ for column-role predicates. The marker rows of group g have entry one against a non-marker column exactly when $\kappa_g(t)$ holds there. Similarly, marker columns of group g have entry one against a non-marker row exactly when $\rho_g(t)$ holds. The corner of a single mode is $J_2 - I_2$, while the corner between different modes is zero. The parity index in the W modes is therefore a genuine structural datum. It separates the two candidate child branches using only one bit, and it is recovered from the parity of the terminal leaf.

Lemma B.3 (Propagation frames realize the prescribed order)**.** *At every depth $1 \le i \le h$ and for every node p_i with parent p_{i-1}, the four row and column pairs displayed in Section 3 are increasing tuples on their respective axes. Their marker restrictions and their incidences with the dummy blocks agree with the top twelve entries of H_4.*

Proof. The first two selected positions are the two groups of one marker mode, both before the variable and dummy positions. The order of parent and child among variable rows is ancestor–descendant for minus and descendant–ancestor for plus; among columns it is the reverse. The first column in a V frame is the column dummy, which precedes the tree. The second row in a W frame is the row dummy, which follows it. Consequently the four increasing row and column tuples are exactly those displayed in the forcing table. The marker corner is the off-diagonal matrix $J_2 - I_2$. Membership in the two row roles gives the lower-left identity, and membership in the two column roles gives the upper-right identity. The dummy incidences are one by the entry definition. The corresponding body entries may change under edits, yielding precisely the implications shown in the forcing table. □

B.2 Where the two signs meet

At depths below h, plus and minus designate different physical row and column blocks even when they carry the same node label. Their diagonal entries can therefore consistently have opposite values. At the leaf level there is only one block $R_h(p)$ and only one block $C_h(p)$; the selected offsets are both zero. Thus the positive and negative inductive requirements apply to the same physical matrix cell, not simply to two equal labels. The identification is the final step required by the edit-distance argument. It also explains why the protected selection must use one common leaf $z = a \oplus b$ on both axes.

C A second proof of the editing lower bound

The exact disjointness of protected sets under the XOR selection is a convenient deterministic proof. A probabilistic argument reaches the same editing threshold using only one-coordinate uniformity. This formulation shows which feature is intrinsic to the tree host and which feature is special to the XOR coupling. The marginal-exposure and union-bound method adapts the edit-distance argument of [6] to the smaller marker frame used here. It is included to explain the construction and to compare it with the deterministic packing, rather than as a new probabilistic method.

Choose a leaf label Z uniformly from $\{0, \dots, m-1\}$. At every depth d select the node $\lfloor Z/2^{h-d} \rfloor$ and, in each signed block at that node, sample one physical row and one physical column uniformly from its 2^{h-d} offsets. Sample the row and column representatives in the marker and dummy blocks uniformly and independently of each other and of the tree representatives. The two signed paths meet at the same selected physical leaf because a leaf block consists of a single position.

Lemma C.1 (Single-position marginal)**.** *For any fixed variable row position at depth d and sign, its probability of selection is m^{-1}. The same is true for a fixed variable column position, marker position, or dummy position.*

Proof. A depth-d node is reached with probability 2^{-d}. Conditional on reaching it, a given offset is sampled with probability $2^{-(h-d)}$. Multiplying gives $2^{-h} = m^{-1}$. At the leaf depth the selection probability is 2^{-h} because there is one representative per node. Marker and dummy blocks have exactly m positions and are sampled uniformly. □

Protect all selected cells for which at least one endpoint is a marker or dummy position, together with the two selected root-variable diagonal cells. No other variable–variable cells are protected. In particular, the shared terminal cell, whose value will be forced by the propagation, need not be protected.

Lemma C.2 (Two-coordinate exposure)**.** *For every fixed physical matrix cell e, the probability that e is protected by the random selection is at most m^{-2}.*

Proof. If either endpoint of e is a marker or dummy position, that endpoint is chosen independently of the tree-path and variable offset choices on the other axis. Each endpoint has marginal probability m^{-1} by Lemma C.1, and the two events are independent, giving m^{-2}. If both endpoints are variable positions at the root, their offsets are sampled independently from the two root blocks of size m, giving the same product. All other cells have protection probability zero. □

Proposition C.3 (Probabilistic lower bound)**.** *Every edit set that turns A_h into an H_4-free binary matrix contains at least m^2 cells.*

Proof. Let E be an arbitrary set of changed cells with $|E| < m^2$. By Lemma C.2 and the union bound,

$$\mathbb{P}(E \cap \text{Protected} \neq \varnothing) \leq \sum_{e \in E} \mathbb{P}(e \in \text{Protected}) \leq |E| m^{-2} < 1.$$

A selection therefore exists whose protected cells avoid every edit. The forcing frames along its common path are valid and have all twelve marker incidences unchanged. The same four implications as in Section 3 propagate the root value one through the plus chain and the root value zero through the minus chain. Both implications concern the same selected terminal cell; they are inconsistent in an H_4-free edited matrix. Thus E cannot eliminate all copies. □

The probabilistic argument uses independence *between axes* whenever one position is in a marker or dummy block. It does not require independence of the variable representatives on different levels of the same axis, nor of the two terminal path labels. The deterministic XOR proof goes further: each individual protected cell identifies at most one of the m^2 selections. This is an exact combinatorial packing statement, while the probabilistic proof controls the probability of hitting each required cell. Both yield the sharp integer threshold m^2.

C.1 Boolean verification of the four implications

The lower-right body P is $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$. Each of the four forcing frames has three of its four body entries fixed by the choice of mode or by the current inductive hypothesis. The remaining entry is constrained by the absence of a completed copy. The four exact incompatibilities are

$$\begin{aligned} V_i^+ &: \quad (x_{i-1}, \alpha_i) = (1, 0), \\ V_i^- &: \quad (y_{i-1}, \beta_i) = (0, 1), \\ W^+_{i,p_i \bmod 2} &: \quad (\alpha_i, x_i) = (1, 0), \\ W^-_{i,p_i \bmod 2} &: \quad (\beta_i, y_i) = (0, 1). \end{aligned}$$

Each forbidden pair would complete the body P and therefore produce H_4 with the protected frame. The complement of each pair yields exactly the implication in the forcing table. This explicit truth-table reduction is independent of the arithmetic classification certificates used for the upper bound.

D Robustness of the edit obstruction under marker permutations

Only a small part of the complete marker order enters the edit-distance argument. The original host places the two groups of each marker mode consecutively and uses a particular ordering of all $6h$ modes. The forcing argument itself requires only that the two groups of a single mode occur in their given order and that the entire marker prefix precede the variable and dummy blocks. This gives a useful separation between the mechanism responsible for distance and the mechanism responsible for terminal localization.

For each depth h, let π_r and π_c be two arbitrary permutations of its $6h$ complete marker modes. Define $A_h^{\pi_r,\pi_c}$ by permuting the marker *row* blocks in accordance with π_r and marker *column* blocks in accordance with π_c, preserving the order of the two groups inside each mode. Keep every block-constant entry, the two signed tree orders, and the positions of the row and column dummy blocks relative to the variable tree. The result is again a square matrix of order $(14h+2)2^h$.

Proposition D.1 (Order-independent forcing lower bound)**.** *For every $h \geq 1$ and every pair of complete-mode permutations,*

$$D_{H_4}(A_h^{\pi_r,\pi_c}) \geq 4^h.$$

In particular, the lower bound holds against arbitrary binary edits under independently reordered marker prefixes.

Proof. Use exactly the m^2 XOR-indexed selections from Section 3. The definition and injectivity of the protected cells depend on physical block labels, their offsets, the chosen tree path and the fact that marker and dummy blocks have size m; permutation of whole marker modes changes none of these. Therefore any edit set of size below m^2 misses the protected cells of one selection.

For a fixed mode t, its two marker row groups appear in the order $A_1(t) < A_2(t)$ and its marker column groups in the order $B_1(t) < B_2(t)$. All four groups precede every variable and dummy position, regardless of the position of other complete modes in the prefix. The four forcing frames use only the two marker groups of their own mode. Their increasing order and all the relevant incidences therefore remain unchanged. The implications in the forcing table again propagate $x_0 = 1$ and $y_0 = 0$ to the common leaf cell, contradicting freeness. Thus any eliminating edit set has size at least m^2. □

The upper bound $D_{H_4}(A_h) \leq m^2$ and the exact copy count were proved for the canonical mode order in Section 2. Proposition D.1 records the broader range of the forcing mechanism itself. The separately supplied Lean extension contains the corresponding concrete-index theorem `permuted_matrix_edit_lower_bound`; the primary submission project retains its canonical order in the global classification and removal statements.

E The selection map as an injective coding device

The XOR choice can be characterized without mentioning the particular marker rules. Fix a word length h, and for each pair $(a, b) \in \{0, \ldots, 2^h - 1\}^2$ place a selected representative in every signed variable block on the ancestral path of $a \oplus b$. At a depth d, the selected node is the high-d-bit prefix of the XOR, and the selected offset on the row axis (respectively column axis) is the low $h - d$ bits of a (respectively b).

Lemma E.1 (One-endpoint recovery)**.** *Suppose the selected variable row at depth d is known and the selected marker-column offset is known. Then the pair (a, b) is uniquely determined. The analogous assertion holds with the two axes reversed. A pair of selected root-variable positions also uniquely determines (a, b).*

Proof. The marker-column offset equals b. The variable row determines its node prefix p_d and row offset $u = a \bmod 2^{h-d}$. Since bitwise XOR commutes with taking high-bit prefixes,

$$\lfloor a/2^{h-d} \rfloor = p_d \oplus \lfloor b/2^{h-d} \rfloor.$$

The quotient on the left and remainder u recover a. This also holds for $d = h$, when the row offset is zero. Reversing the axes recovers b by the same equation. At $d = 0$, the root node is zero and the row and column offsets are exactly a and b. □

This coding interpretation makes the protected-cell injectivity transparent. Each protected cell contains at least one such recoverable configuration; hence it belongs to at most one selected path. The proof uses all m^2 labels simultaneously. Independent choices of a random leaf and unrelated row and column offsets would not provide this exact disjointness, even though some marginal probabilities would still be uniform. The deterministic injection also eliminates the need to estimate correlations among the successive forcing gadgets.

F Which cells the canonical repair changes in an original copy

The two terminal families give a precise description of the intersection of each original copy with the canonical edit set

$$S_h = R_* \times \bigcup_{0 \le p < m} C_h(p).$$

Here $|S_h| = m^2$ is a count of physical cells. The classification reveals that the same repair affects the two terminal forms with different multiplicities.

Proposition F.1 (Repair-incidence spectrum)**.** *Every type-I physical copy of H_4 in A_h contains exactly one cell from S_h. Every type-II copy contains exactly two cells from S_h. The number of original copies meeting S_h in one cell is $2m^6$, and the number meeting it in two cells is $\frac{1}{2}(m-2)m^5$.*

Proof. The unique row type belonging to R_* in either terminal form is the fourth row. In type I, the only chosen leaf column is the third column $C_h(p)$; the two first columns are markers and the fourth is the signed parent. Therefore precisely the fourth-row, third-column physical cell belongs to S_h. In type II, both the second and third columns are leaves, namely $C_h(q)$ and $C_h(p)$. They are distinct since $q < p$. The first column is a marker and the fourth is a signed parent. Thus exactly two cells of the copy belong to S_h. The counting assertion follows from the exhaustive, disjoint terminal classification and the weighted counts in Section 5. □

Corollary F.2 (The canonical rectangle hits every old copy)**.** *Every copy of H_4 present in A_h has at least one entry changed in the passage to B_h.*

Proof. Both terminal families meet S_h by Proposition F.1, and there are no other original copies. □

There are two distinct assertions in the proof of the sharp distance. The corollary identifies a small set intersecting the originally present copies; Lemma 4.2 proves that after all cells of this set are changed simultaneously, the resulting physical matrix is completely H_4-free. The latter requires the repaired entry-table classification, because a cell edit can participate in configurations absent from the original matrix. The arithmetic repair certificate establishes freeness directly for the edited matrix, rather than inferring it from the hitting corollary.

The spectrum also provides a quantitative accounting check. If one sums, over original physical copies, the number of their entries inside S_h, then the answer is

$$2m^6 + 2 \cdot \frac{m-2}{2} m^5 = (3m-2)m^5. \tag{7}$$

This number counts copy–edit-cell incidences with multiplicity; it is deliberately different from both the number of copies $\left(\frac{5}{2} - \frac{1}{m}\right) m^6$ and the cardinality m^2 of the editing rectangle. Equation (7) can be verified independently from the terminal parameterization.

G Arithmetic certificates and formalization

This appendix specifies the integer model used in the classification and repair proofs. Its domain and entry rules are uniform in h, so the two certificate lemmas have a single depth parameter.

G.1 The integer block domain

For an integer $h \geq 1$, the admissible records are of three kinds:

$$\begin{aligned} M(j,i,g) &= (0,j,i,g,0,0,0), \\ U(d,p,s) &= (1,0,0,0,d,p,s), \\ D_* &= (2,0,0,0,0,0,0). \end{aligned}$$

The seven fields are, in order, kind, mode, marker level, marker group, variable depth, node, and sign. A marker satisfies $0 \leq j < 6$, $1 \leq i \leq h$ and $g \in \{1,2\}$. The mode codes are

$$0 = V^+, \quad 1 = V^-, \quad 2 = W_0^+, \quad 3 = W_1^+, \quad 4 = W_0^-, \quad 5 = W_1^-.$$

A variable satisfies $0 \leq d \leq h$, $p \geq 0$, and $d = 0 \Rightarrow p = 0$. Its sign is $s = 2$ when $d = h$, and $s \in \{0,1\}$ when $d < h$, with $0, 1, 2$ denoting plus, minus and leaf, respectively.

The physical encoding assigns the appropriate mode code to a marker, copies the depth, node and sign of a variable, and sends the dummy to D_*. Thus every physical block gives an admissible record. The integer domain relaxes physical validity by omitting the bound $p < 2^d$.

G.2 Comparison and entry predicates

For markers, comparison is lexicographic on

$$(\mathbf{1}[j < 4], i, j, g).$$

This places modes 4 and 5 first, in precisely the marker order of Section 2. Every marker precedes every non-marker. Among non-markers the dummy follows variables on rows and precedes them on columns.

For variable records, write $(o_r, c_r) = (1,0)$ and $(o_c, c_c) = (0,1)$ for the opening and closing signs on the two axes. Put $q' = \lfloor q/2 \rfloor$ and $p' = \lfloor p/2 \rfloor$. The integer comparison $\prec_a$, for $a \in \{r, c\}$, is defined by

$$U(d,p,s) \prec_a U(e,q,t) \iff \begin{cases} p < q \ \vee \ (p = q \wedge s = o_a \wedge t = c_a), & d = e, \\ p < q' \ \vee \ (p = q' \wedge s = o_a), & e = d+1, \\ p' < q \ \vee \ (p' = q \wedge t = c_a), & d = e+1, \\ \text{true}, & |d - e| \geq 2. \end{cases} \tag{8}$$

At equal depths and adjacent depths these are the exact comparisons of the tree blocks. At larger depth gaps the relation permits every pair.

Let E_h^0 be the original integer entry function. Its marker–marker rule is

$$E_h^0(M(j,i,g), M(k,\ell,u)) = \mathbf{1}[j = k \wedge i = \ell \wedge g \neq u].$$

The marker roles are those of the table in Section 2, with level and parity tested on the records. In a signed role, $U(d,p,2)$ belongs to either sign when $d = h$. Entries with a dummy and a non-marker are one. The variable entries are exactly (4), with the depth and sign conventions above. Let E_h^1 have the same rules, except

$$E_h^1(D_*, U(h,p,2)) = 0 \qquad (p \geq 0).$$

These definitions retain every entry rule while relaxing node ranges and some comparisons.

For $\eta \in \{0,1\}$, define $\mathcal{C}_h^\eta(r,c)$ for two four-tuples of records by the following conditions: all eight records are admissible, $r_i \prec_r r_j$ and $c_i \prec_c c_j$ for every $1 \leq i < j \leq 4$, and

$$E_h^\eta(r_i, c_j) = (H_4)_{ij} \qquad (1 \leq i, j \leq 4).$$

In particular, all six comparisons on each axis are part of the copy predicate.

Lemma G.1 (Encoding a physical copy)**.** *The physical block encoding maps every ordered block copy in A_h to a witness for $\mathcal{C}_h^0$, and every ordered block copy in B_h to a witness for $\mathcal{C}_h^1$.*

Proof. Physical marker levels and groups satisfy the integer ranges. For a variable, physical validity gives its depth and sign, the nonnegative node label, and the root condition. Thus the encoding preserves validity.

For comparisons, a marker pair has the same lexicographic key on both axes. A marker–non-marker pair and a dummy–variable pair follow the stated axis rules. For two variables at equal depth, node order and the opening and closing signs give the first clause of (8). At adjacent depths the ancestor node is the integer quotient by two, giving the second and third clauses. At a gap of at least two levels the relaxed clause is satisfied. Hence all physical comparisons imply their integer counterparts, separately for rows and columns.

Finally, marker pairs use the same mode and group test, mixed marker pairs use the same role test, and the non-marker entries use the same dummy and variable rules. The canonical change is precisely on the encoded dummy-row–leaf-column products. All sixteen entry equalities are therefore preserved. These arguments are the encoding lemmas `encode_valid`, `encode_before`, `encode_original_entry` and `encode_repaired_entry`; together they give `encode_copy`. □

G.3 The certificate statements

Define $T_h(r, c)$ to mean that there exist $j \in \{4, 5\}$ and $p \geq 0$ such that

$$\begin{aligned} r &= (M(j, h, 1), M(j, h, 2), U(h, p, 2), D_*), \\ c &= (M(j, h, 1), v, U(h, p, 2), U(h - 1, \lfloor p/2 \rfloor, 1)), \end{aligned}$$

where either $v = M(j, h, 2)$, or $v = U(h, q, 2)$ with $0 \leq q < p$ and $q \equiv p \pmod 2$. The marker parity is recovered from the entry conditions when this predicate is decoded in the physical host.

Lemma G.2 (Arithmetic classification and repair)**.** *For every integer $h \geq 1$ and every pair of four-tuples r, c of integer records,*

$$\mathcal{C}_h^0(r, c) \implies T_h(r, c), \qquad \mathcal{C}_h^1(r, c) \implies \bot. \tag{9}$$

Certificate proof. The first implication is the Lean theorem `Arithmetic.classification`, and the second is `Arithmetic.repaired_no_copy`. The classification derivation has 266 source segments and the repair derivation has 243. Their steps are propositional inferences, linear integer arithmetic lemmas, and congruence proofs. The transparent formula abbreviations express the admissibility, comparison and entry predicates specified above.

The two bridge modules instantiate the final derivations with the fields of r and c. Both copy predicates supply eight validity conditions, twelve pairwise comparisons and all sixteen entry conditions. The classification derivation needs only fifteen of the entry conditions: it does not use the condition at $(0, 0)$ (with zero-based indices). The repair derivation uses all sixteen. For classification the negation of T_h gives the counterexample premise; for repair the premises are the repaired copy conditions themselves. Thus the omitted classification premise strengthens that certificate without weakening the definition of an induced copy. The resulting contradictions establish (9). The derivations and the bridge proofs are checked by the Lean kernel. □

To decode the first conclusion, the leaf-row entry against the first marker column is one. The role test consequently gives $j = 4 + (p \bmod 2)$. Physical validity then supplies $p < 2^h$ and the valid minus parent. This is the theorem `terminal_decodes`, which yields the terminal families in Section 4.

G.4 Formal statements and verification

The matrix family `standardHost` uses the usual order on $\mathrm{Fin}(n_h)$ on both axes. Its copy predicate uses increasing four-tuples and all sixteen Boolean equalities. Its edit distance is the attained minimum Hamming cost over pattern-free binary matrices. The principal mathematical interfaces are

Result	Lean declaration
Physical size and order	`physicalPairs_length`, `orderedPosition_order_iff`
Matrix copy bridge	`standard_copy_iff`
Terminal classification	`classification_iff`
Repaired freeness	`canonical_standard_free`
Exact copy count	`exact_copy_count`
Arbitrary-edit lower bound	`arbitrary_edit_lower_bound`
Exact edit distance	`exact_edit_distance`, `exact_normalized_distance`
Asymptotic ratio	`normalized_copy_ratio_tendsto`
Removal obstruction	`no_polynomial_removal_bound`
Main theorem	`end_to_end_main_theorem`

The XOR construction is developed in `XorSampling.lean`; its counting argument is in `FiniteAvoidance.lean`. Terminal parameters are counted in `TerminalCount.lean`, and their physical realizations are enumerated in `CopyFibres.lean`. The module `GlobalCounting.lean` identifies their disjoint union with the full copy enumeration. `EditDistance.lean` transports Hamming cost through the order-preserving bijections and proves attainment of the minimum.

The project pins Lean 4.24.0 and Mathlib at revision

`f897ebcf72cd16f89ab4577d0c826cd14afaafc7`.

The clean verification procedure is run from the project root by

```
bash verify_all.sh
```

Its declaration audit selects every definition and theorem by its defining project module, including private and anonymous declarations, and checks the complete axiom dependency closure. Only `propext`, `Classical.choice` and `Quot.sound` are allowed. Current run records, exact source and artifact hashes, and dependency provenance are under `evidence/strict_build/`; supplied older logs are preserved separately as historical evidence. Appendix I describes the build procedure, proof dependencies and retained verification records.

H A uniform finite-witness principle for the host family

The integer certificate in Appendix G handles arbitrary depths directly. The same host also admits a finite-depth reduction that is useful for independent specification checks. We give its combinatorial proof in the notation of the present paper. The statement applies to any square pattern whose row signatures and column signatures are individually distinct. This structural result is logically separate from the arithmetic certificate proof of Theorem 1.1.

Lemma H.1 (Number of binary branching levels)**.** *For a family of $v \geq 1$ finite binary words, at most $v - 1$ different bit positions are the first divergence position of some pair before either word terminates.*

Proof. Form the trie of the given words, regarding a word that is a proper prefix of another as a marked internal vertex. A first divergence occurs only at a vertex with two nonempty child branches. A rooted tree with v distinct marked word endpoints has at most $v - 1$ such branching vertices: each branching vertex separates at least two components containing marked

endpoints, and contracting all degree-one vertices yields a rooted tree with at most v leaves after marked internal vertices are represented as terminal branches. Consequently the number of distinct branching depths is at most $v - 1$. □

Lemma H.2 (Retained-level preservation)**.** *Choose at most $2k$ row or column quotient types from the host at height h. Let $S \subseteq \{0, \dots, h\}$ contain $0, h$, the defining depth of each selected marker or variable type, the immediately preceding depth of each such positive defining depth, and every first-divergence level of a pair of selected variable words. Write $S = \{s_0 = 0 < s_1 < \cdots < s_{h'} = h\}$. Replace every selected defining depth s_i by i and every selected node word by its bits in retained positive positions. Then the compressed types preserve all required block comparisons and entries of both the original host and its canonical repair.*

Proof. The rank map preserves comparisons and equality of selected depths. The predecessor of every selected positive depth is retained. Therefore two selected depths are consecutive precisely when their images are consecutive; a gap of length at least two retains a point strictly between the endpoints. The first-divergence bit of every pair is retained, so lexicographic order and the proper-prefix relation are preserved. Since the final bit at any defining variable depth is retained, so is its parity. Parent comparison at adjacent depths is preserved because the relevant predecessor bit is retained.

These are exactly the relations appearing in the host rules: marker pairs test equality of mode and inequality of group; marker–variable pairs test family, sign, depth adjacency, and parity; variable pairs test equal or adjacent depths with their lexicographic or parent comparison. The dummy incidence depends only on type; the repair tests dummy row against a terminal leaf column. Every selected type survives, its sign and marker group are unchanged, and these Boolean tests have the same values after compression. Lemma B.1 gives preservation of the signed tree order. The marker mode order depends on family, retained depth and group, so its relevant comparisons are preserved as well. For equal node words at an internal level, the two signs remain distinct and their opening-before-closing order is unchanged. □

Theorem H.3 (General $6k$-level small-witness theorem)**.** *Fix $k \geq 1$ and a binary $k \times k$ matrix F with pairwise distinct rows and pairwise distinct columns. If F occurs as an induced ordered copy in A_h or B_h for some $h \geq 1$, then it occurs in the corresponding host at some depth $1 \leq d \leq \min\{h, 6k\}$. Consequently, the whole original (resp. repaired) host family is F-free if and only if its depths $1, \dots, 6k$ are all F-free.*

Proof. By Proposition B.2, the given physical copy projects to $2k$ types, distinct among the selected row types and among the selected column types. Retain their defining levels and predecessors, as well as $0, h$ and every first-divergence bit level of the selected variable words. If $v \geq 1$ is the number of variable types, Lemma H.1 bounds the number of first-divergence levels by $v - 1 \leq 2k - 1$. Thus

$$|S| \leq 2 + 2k + 2k + (2k - 1) = 6k + 1.$$

If $v = 0$, the still stronger bound $|S| \leq 2 + 4k$ holds. Since both endpoints $0, h$ are in S, the reduced height $d = |S| - 1$ satisfies $1 \leq d \leq \min\{h, 6k\}$. By Lemma H.2, the compressed types retain each of the k^2 prescribed entries and the two increasing orders. They remain distinct within either axis: different defining kinds, levels or signs remain different, and distinct node words remain different because their first divergence is retained. Every compressed block has positive multiplicity, so representatives of the new types furnish a physical copy of F at depth d. The converse direction of the final equivalence is immediate; if some larger depth contained a copy, the first assertion would supply a copy at one of the finitely many depths. □

Corollary H.4 (Twenty-four levels suffice for a copy witness)**.** *Any H_4 copy in A_h or B_h at arbitrary depth yields an H_4 copy in a host of the same kind at depth at most 24.*

Proof. Apply Theorem H.3 with $k = 4$. □

Remark H.5 (Proof scope)**.** *The finite-witness theorem is available in the separately supplied Lean development in the declarations* `physical_pattern_copy_has_small_witness` *and* `matrix_pattern_copy_has_small_witness`*. These exported interfaces certify* $1 \le d \le 6k$*. The additional bound* $d \le h$ *in the paper statement follows from* $S \subseteq \{0, \dots, h\}$ *in the displayed combinatorial construction; it is not part of those exported Lean conclusions. The main theorem of this article uses the arithmetic-classification and arithmetic-repair route in the primary formal source release.*

H.1 Finite decision and its precise meaning

Theorem H.3 supplies an exact finite decision procedure for the assertion that *every depth* of the specified host family is F-free. For each of the $6k$ allowed depths, enumerate ordered quotient tuples with k distinct row and k distinct column types and evaluate all k^2 entries. Absence at all these depths implies absence in every larger host. The statement concerns a fixed explicit family of hosts, rather than arbitrary $n \times n$ matrices; its finite reduction does not impose a polynomial running-time bound in k. For H_4, the arithmetic approach provides a direct all-depth proof without the need to run the finite enumerator.

I Reproducible verification and the proof interfaces

The formalization has four components: physical block positions and their multiplicities; the ordered quotient host; arithmetic classification and repaired freeness; and transfer to matrices indexed by $\mathrm{Fin}(n_h)$. The theorem `end_to_end_main_theorem` combines these components with the exact-count, arbitrary-edit distance and asymptotic results.

I.1 Dependencies of the principal statements

Mathematical assertion	Lean declaration or module	Role
Arithmetic original classification	`Arithmetic.classification`	Copy exhaustion
Repaired freeness	`Arithmetic.repaired_no_copy`	No new copies
Encoding of physical copies	`Arithmetic.encode_copy`	Arithmetic soundness
Terminal decoding	`terminal_decodes`	Physical terminal forms
Numeric matrix order	`orderedPosition_order_iff`	Order transfer
Protected sampling	`XorSampling.lean`	Arbitrary edits
Finite avoidance	`FiniteAvoidance.lean`	Untouched witness
Copy multiplicity	`CopyFibres`, `TerminalCount`	Exact count
Edit distance	`EditDistance.lean`	Exact minimum
Asymptotics	`Limits.lean`	Removal failure

The arithmetic classification and repair derivations contain 266 and 243 source segments, respectively. The encoding checks all eight validity conditions, twelve axis-order comparisons and sixteen pattern entries. The quotient-to-physical correspondence and the numeric-order bijections complete the passage to the ordinary indexed matrix. The proof segments and every imported Lean module are distributed as source files in the accompanying verification archive.

I.2 Rebuild and axiom audit

The project uses Lean 4.24.0 and pins Mathlib to revision

`f897ebcf72cd16f89ab4577d0c826cd14afaafc7`.

From the `formal_verification/` directory, execute

```
bash verify_all.sh
```

The script retrieves the pinned dependency cache, builds the primary library, checks the main theorem and its axiom audit, then rebuilds the isolated structural-extension and exact-transversal projects. The main library is complete with all arithmetic certificates, which are needed for the terminal classification and canonical freeness results. The compiled dependency audits permit only `propext`, `Classical.choice` and `Quot.sound`.

The archived primary build record reports a compilation of 550 project modules, with 116,338 declarations (93,189 theorems) in the strict declaration audit. The independent extension-core build record covers 34 library modules and 1,570 declarations (991 theorems). The exact-transversal supplement records 49 declarations, including 32 theorems, and derives terminal exhaustion from the primary global classification theorem. The corresponding run reports and source identities are retained under `evidence/` and in the supplemental project directories. The archived records identify the source and environment of earlier compilations; the command above performs a new build on the recipient's machine.

I.3 Mathematical cross-checks

A direct check of the construction first verifies block-constant entries and the four increasing forcing frames. The encoding implication then maps every physical copy to the arithmetic host. The classification certificate excludes all nonterminal copies; the repair certificate excludes every copy after the canonical edit. Finally, weighted terminal multiplicities give the exact number of original copies. The main Lean theorem assembles these facts with the arbitrary-edit lower bound and the asymptotic comparison.

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